Healthcare Risk Prediction — Machine Learning vs Neural Networks

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1. Problem Understanding

Description of the Problem

This project utilizes a healthcare dataset containing 6,000 patient records with various health indicators to build predictive models that can identify individuals at risk of diabetes.

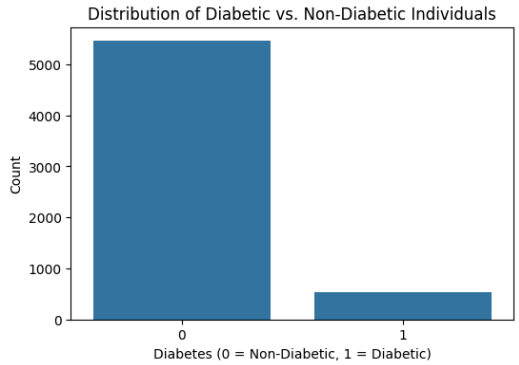
Importance of Predicting Diabetes Risk

Early identification of diabetes risk is critical for healthcare providers to implement preventative measures, such as lifestyle interventions and pharmacological treatments. Accurate prediction allows for the prioritization of clinical resources for those most in need, potentially reducing the long-term healthcare burden and improving patient quality of life.

2. Data Exploration Insights

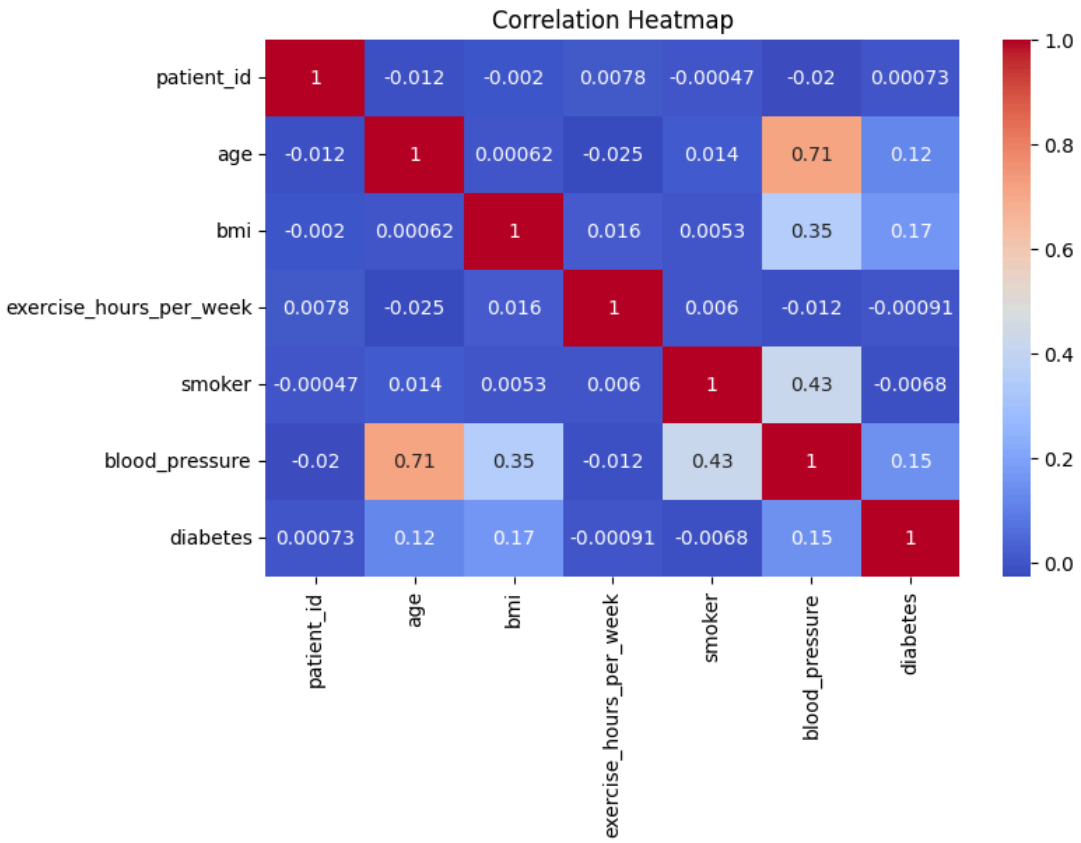
Key Observations from the Dataset

- **Dataset Composition**: The dataset consists of 6,000 rows and 7 columns.
- **Features**: The data includes `patient_id`, `age`, `bmi`, `exercise_hours_per_week`, `smoker`, `blood_pressure`, and the target variable `diabetes`.
- **Data Integrity**: There are no missing values in any of the columns.
- **Class Imbalance**: Approximately 9% of the individuals in the dataset are diabetic (target = 1), indicating a significant class imbalance.



Important Features Influencing Prediction

- **Age(0.12)**: Analysis shows that individuals with diabetes have a higher median age (~60 years) compared to non-diabetic individuals (~50 years).
- **BMI (0.17)**: High BMI is a notable predictor, with diabetic patients showing a distribution skewed toward higher values.
- **Blood pressure(0.15)**: Preliminary visualizations suggest that lower weekly exercise hours correlate with a higher risk of being diabetic.



3. Model Implementation

Machine Learning Model: Logistic Regression

The first model implemented was a **Logistic Regression** classifier.

- **Preprocessing**: Features were separated from the target variable, followed by a train-test split and feature scaling using `StandardScaler`.
- **Handling Imbalance**: To address the 91/9 split, the model used the `class_weight='balanced'` parameter, which automatically adjusts weights inversely proportional to class frequencies.

```
              0                       1
           precision    recall  f1-score   support

        0       0.96      0.62      0.75      1091
        1       0.16      0.72      0.26       109

 accuracy                           0.63      1200
macro avg       0.56      0.67      0.51      1200
weighted avg    0.88      0.63      0.71      1200
```

Neural Network Architecture

A **Sequential Artificial Neural Network (ANN)** was built using TensorFlow/Keras.

- **Preprocessing**: To handle class imbalance for the ANN, **SMOTE** (Synthetic Minority Over-sampling Technique) was applied to the training data.
- **Layers**: The architecture includes:
  - An input layer.
  - Hidden layers using the **ReLU** activation function.
  - An output layer with a **Sigmoid** activation function for binary probability output.
- **Compilation**: The model was compiled using the **Adam optimizer** and **binary cross-entropy loss**.

4. Model Evaluation

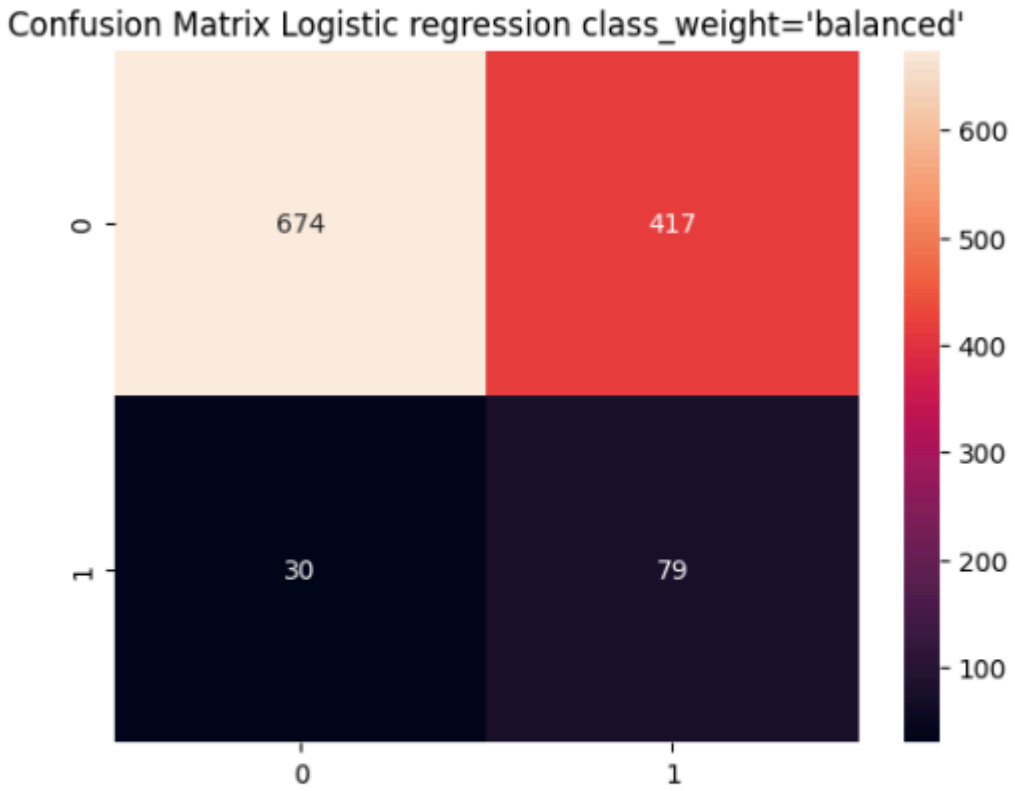
Performance Metrics

Metric	Logistic Regression (Balanced)	Neural Network (with SMOTE)

Accuracy	Generally lower due to higher false positives.	Higher overall accuracy.
Recall (Diabetes)	**Higher**: Very effective at catching diabetic cases.	Significant improvement after SMOTE.
F1 Score	Balanced for the minority class.	Optimized for overall performance.

Visualizations

Confusion Matrix: Used to interpret the trade-off between sensitivity and specificity.



	precision	recall	f1-score	support
0	0.96	0.62	0.75	1091
1	0.16	0.72	0.26	109
accuracy			0.63	1200
macro avg	0.56	0.67	0.51	1200
weighted avg	0.88	0.63	0.71	1200

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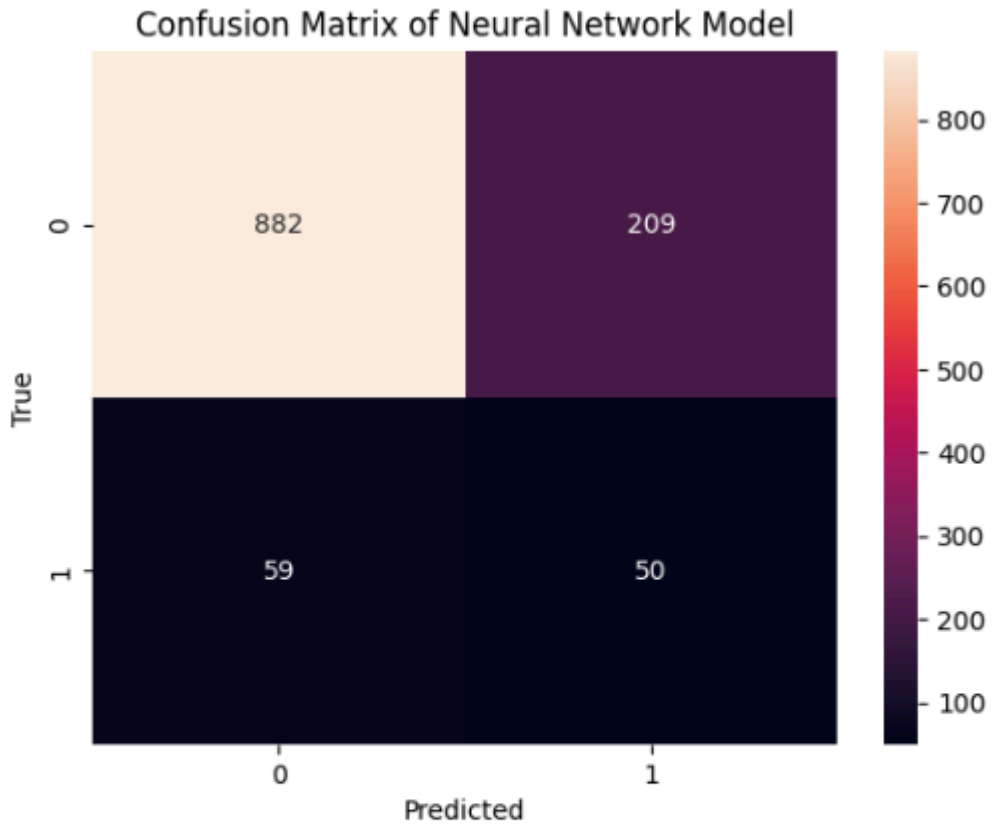
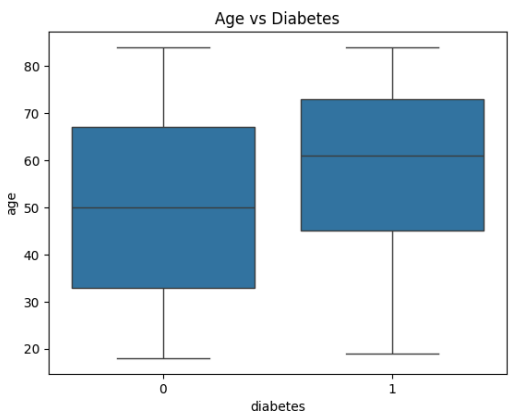
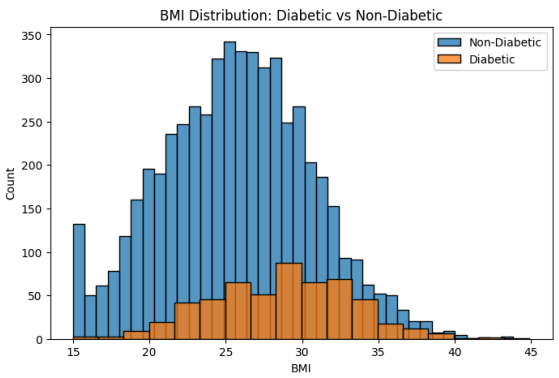


	precision	recall	f1-score	support
0	0.94	0.81	0.87	1091
1	0.19	0.46	0.27	109
accuracy			0.78	1200
macro avg	0.57	0.63	0.57	1200
weighted avg	0.87	0.78	0.81	1200

Other Graphs





5.Comparitive Analysis

Logistic Regression (balanced) might be preferred when:

- The primary goal is to maximize recall for the minority class (e.g., detecting a rare disease), where missing a positive case is highly undesirable, and initial results suggest it achieves higher minority class recall.
- Model interpretability is important for understanding the contributing factors.
- Computational resources are limited.

Neural Networks (with SMOTE) might be considered when:

- A balance between overall accuracy and minority class performance is desired, and the model is capable of capturing complex non-linear patterns.
- Further fine-tuning (e.g., hyperparameter tuning, different architectures) can potentially lead to even better performance for both classes.

Overall, both models show different strengths. The Logistic Regression with `class_weight='balanced'` still has a higher recall for the diabetic class, making it potentially more suitable if minimizing false negatives for diabetes is the absolute priority. However, the Neural Network with SMOTE shows a better overall accuracy and a significant improvement in identifying diabetic patients, suggesting it could be a strong contender with further optimization.